Remote Environmental Survey and Mapping of an Unseen Room

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1. Problem and User Need

Problem. Before entering an unfamiliar enclosed room, a person may not know the temperature, humidity, potential gas hazards, or layout of the room. This creates a risk of entering an unsafe environment without sufficient information.

Needs (stakeholder 1: a technician inspecting a room)

- **N1** "I need to see the room layout and identify where potential hazards are located."
- **N2** "I need to explore an unfamiliar room without physically entering it."
- **N3** "I need to understand how environmental conditions vary across different areas of the room."

Needs (stakeholder 2: an emergency responder)

- **N4** "I need to access the collected information while moving around without relying on a computer."
- **N5** "I need to see what is happening inside the room so I can navigate around obstacles remotely."
- **N6** "I need to identify potentially hazardous areas before entering to reduce unnecessary exposure."

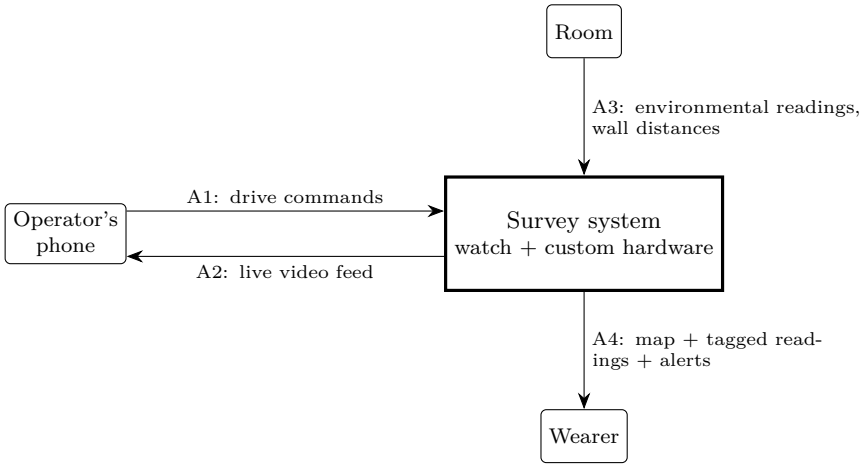
Needs (both stakeholders)

- **N7** "I need the system to stay within the available budget."
- **N8** "I need the system to be documented so another person can rebuild or maintain it."

Objective. In a single survey run, produce a map of a 4 m × 4 m room that identifies at least three measurement locations with corresponding temperature, humidity, and gas readings, without requiring the operator to enter the room.

2. Scope and System Context

In scope	Out of scope (neighbour)
Remote operation from a phone with live video feed	Autonomous navigation and automatic route planning
A 2D map of one room with measurement locations	3D mapping, multiple rooms and stairs
Temperature, humidity and relative gas measurements	Identifying gas species, locating gas sources or certifying room safety
A hazard-indicated survey summary on the watch	Cloud storage and long-term monitoring
A low-fidelity prototype on breadboard/perfboard	PCB design and manufacture (Project II)



Stakeholder collision. The technician (N1, N3) wants a detailed room map and readings from several locations, which takes survey time. The emergency responder (N6) wants potential hazard information quickly, even if the survey is incomplete. Longer measurements improve detail but delay the first useful result.

3. Platform and Custom Hardware

Watch constraints (vendor wiki/product page; datasheet confirmation pending):

- **C1** ESP32-S3R8, up to 240 MHz. Constraint.
- **C2** 2.4 GHz Wi-Fi b/g/n and Bluetooth 5 LE. Constraint; the product page also lists ax, so the datasheet will decide which to cite.
- **C3** Battery 400 mAh at 3.7 V nominal. Constraint.
- **C4** Display 2.06 in AMOLED, 410×502. Constraint.
- **C5** No built-in motor driver. Constraint.

Custom hardware (low fidelity). A two-wheel mobile unit uses a motor driver, controller board, camera, distance sensor and wheel encoders for mapping, temperature/humidity sensor, relative gas sensor and battery. Modules are wired on a breadboard/perfboard. The unit is remotely driven from a phone, collects readings and produces a room map. Its survey summary is sent wirelessly to the watch (C2), which shows the results (C4). The watch's I2C, UART and USB pads are not used because this is a wireless link; the custom hardware remains physically separate.

4. Requirements

Each acceptance target below is selected before testing. These targets and their assumptions are recorded in the logbook.

ID	Class	Statement	Reason	Method: acceptance	Need
REQ-RS-01	Functional	The unit shall complete a 2 m remotely commanded path with the operator outside the room.	2 m provides a basic remote-entry demonstration.	Demonstration: pass if a 2 m path is completed using phone commands only.	N2
REQ-RS-02	Interface	The system shall display live camera video on the operator's phone with latency at most 1 s.	Limit for responding to visible obstacles.	Test: 5 doorway trials; pass if measured video delay is at most 1 s in each.	N5
REQ-RS-03	Functional	The map shall show at least 3 location-tagged records, each containing temperature, humidity and relative gas readings.	Three locations demonstrate readings from different areas of one room.	Demonstration: pass if 3 distinct map markers each display all 3 readings.	N3
REQ-RS-04	Functional	The map shall flag a location as a potential gas anomaly when its test reading is at least 125% of baseline.	A 25% rise is a repeatable *relative* indicator, not a safety limit.	Test: simulated baseline 100; pass if 125 is flagged and 124 is not.	N6
REQ-RS-05	Performance	Mapped wall positions shall be within ±10 cm of tape-measured positions in at least 4 of 5 runs.	10 cm is half the assumed 20 cm unit width.	Test: 8 wall points per run; pass if all 8 meet the bound in at least 4 of 5 runs.	N1
REQ-RS-06	Interface	The watch shall receive the survey summary within 5 s of survey completion.	The wearer needs access at the doorway without a desk setup.	Test: 5 timed runs; pass if every summary arrives within 5 s.	N4
REQ-RS-07	Constraint	Total hardware cost shall not exceed CAD $700.	Limit set by the actual available budget (N7).	Inspection: pass if itemised hardware receipts do not exceed CAD $700.	N7
REQ-RS-08	Buildability	The unit's firmware shall build from a clean copy using one documented command.	Another person must be able to reproduce the build.	Demonstration: pass if a second computer builds it without source edits.	N8

5. Uncertainties and PoC Plan

R1 (largest). *Statement:* wheel slip and distance-sensor noise may cause wall-position error above 10 cm. *Assumption:* odometry and ranging meet REQ-RS-05 in a 4 m × 4 m room. *Consequence:* room boundaries and measurement positions may be misleading (N1). *Probability:* unknown. *Evidence:* map-to-tape comparisons across repeated runs. *Mitigation:* reduce speed, use additional reference points or change the mapping method.

R2. *Statement:* gas readings may drift with humidity or respond too slowly. *Assumption:* a change can be detected during a brief stop. *Consequence:* potential anomalies may be missed (N6). *Probability:* unknown. *Evidence:* repeated safe test exposures and baseline checks at two humidity levels. *Mitigation:* increase dwell time or change sensor. This is not certified gas safety detection.

R3. *Statement:* the wireless link may fail through the doorway. *Assumption:* video reaches the phone and the survey summary reaches the watch from the test room. *Consequence:* remote viewing or portable results are interrupted (N4, N5). *Probability:* unknown. *Evidence:* link tests at the doorway and 3 m inside. *Mitigation:* reduce data rate, buffer the survey and resend after reconnection.

PoC plan (R1 only). *Question:* can the mobile unit map wall locations within the 10 cm bound? *Measure:* absolute wall-position error (cm) at 8 tape-measured points in a 4 m × 4 m area, repeated over 5 runs. *Pass (fixed now):* in at least 4 of 5 runs, every one of the 8 points differs from the reference by at most 10 cm. No measurements or results appear in this proposal.

6. Assumptions and Logbook

- A flat, hard-floored 4 m × 4 m test room and tape-measured wall references are available.
- The mobile unit is approximately 20 cm wide (assumption behind the 10 cm mapping target).
- The watch can receive the survey summary wirelessly; the phone can receive live video feed outside the doorway.
- The gas sensor reports relative changes only; a rise in its output does not establish that air is safe or unsafe.
- The available hardware budget is CAD $700.

AI use. An AI assistant helped revise wording, requirement traceability and test planning. I checked everything the AI wrote against my own requirements, calculations and test plans, and recorded that review in logbook entry L-01.

Key numerical targets and assumptions trace to dated logbook entries L-01.